B. Lunatic Never Content

time limit per test 2 seconds
memory limit per test 256 megabytes
input standard input
output standard output

Rating: 1100 | Tags: math, number theory | https://codeforces.com/problemset/problem/1826/B

You have an array a of n non-negative integers. Let's define $f(a, x) = [a_1 \bmod x, a_2 \bmod x, \ldots, a_n \bmod x]$ for some positive integer x. Find the biggest x, such that $f(a, x)$ is a palindrome.

Here, $a \bmod x$ is the remainder of the integer division of a by x.

An array is a palindrome if it reads the same backward as forward. More formally, an array a of length n is a palindrome if for every i ($1 \le i \le n$) $a_i = a_{n - i + 1}$.

Input

The first line contains a single integer t ($1 \le t \le 10^5$) — the number of test cases.

The first line of each test case contains a single integer n ($1 \le n \le 10^5$).

The second line of each test case contains n integers a_i ($0 \le a_i \le 10^9$).

It's guaranteed that the sum of all n does not exceed 10^5.

Output

For each test case output the biggest x, such that $f(a, x)$ is a palindrome. If x can be infinitely large, output 0 instead.

Example

Input

```
4
2
1 2
8
3 0 1 2 0 3 2 1
1
0
3
100 1 1000000000
```

Output

```
1
2
0
999999900
```

Note

In the first example, $f(a, x = 1) = [0, 0]$ which is a palindrome.

In the second example, $f(a, x = 2) = [1, 0, 1, 0, 0, 1, 0, 1]$ which is a palindrome.

It can be proven that in the first two examples, no larger x satisfies the condition.

In the third example, $f(a, x) = [0]$ for any x, so we can choose it infinitely large, so the answer is 0.